

Python Tuples: A Reference

A tuple in Python is an ordered, **immutable** collection of items enclosed in parentheses (`()`). Unlike lists, tuples cannot be changed, added to, or reordered once created. They are ideal for data that must remain constant throughout your program.

1. Basic Core Mechanics

Tool / Feature	Instruction & Core Mechanics	Step-by-Step Code & Output
<code>(item1, item2)</code> (Creation)	Defines a tuple using parentheses. Note: Creating a single-item tuple requires a trailing comma (e.g., <code>(5,)</code>), otherwise Python treats it as an integer in brackets.	<pre>>>> point = (10, 20) >>> type(point) <class 'tuple'> >>> single = (5,) # Trailing comma needed >>> type(single) <class 'tuple'></pre>
<code>len()</code>	Counts the total number of items stored inside the tuple.	<pre>>>> status_codes = (200, 404, 500) >>> len(status_codes) 3</pre>
<code>in</code>	Checks whether an item exists inside the tuple. Returns <code>True</code> or <code>False</code> .	<pre>>>> roles = ("admin", "user") >>> "admin" in roles True >>> "guest" in roles False</pre>

2. Immutability (The "Lock" Feature)

Action Tried	Instruction & Core Mechanics	Step-by-Step Code & Output
Item Assignment (Why it errors)	Attempting to overwrite, modify, or add an item to a tuple after creation will cause Python to immediately throw a <code>TypeError</code> .	<pre>>>> info = ("London", "UK") >>> info[0] = "Paris" Traceback (most recent call last): File "<stdin>", line 1, in <module> TypeError: 'tuple' object does not support item assignment</pre>

3. Reading & Extracting Data

Method / Syntax	Instruction & Core Mechanics	Step-by-Step Code & Output
<code>.index(val)</code>	Searches the tuple and returns the index position where the specified value first appears.	<pre>>>> response = ("success", 200, "OK") >>> response.index(200) 1</pre>
<code>.count(val)</code>	Scans the entire tuple and returns how many times a value appears.	<pre>>>> track = ("yes", "no", "yes") >>> track.count("yes") 2</pre>
Tuple Unpacking	Extracts all values from a tuple directly into individual variables in a single operation. The number of variables must match the length of the tuple.	<pre>>>> coords = (51.5074, -0.1278) >>> lat, lon = coords # Unpacks values >>> lat 51.5074 >>> lon -0.1278</pre>